

ASSIGNMENT #04

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Due Date: Friday, March 20th TA's Mailbox

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CHE-301 Assignment-04

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Contents

1	Abstract	3
2	Problem A	4
2.1	Derivation of Nitrogen Data and Tables	4
2.2	Problem A, Part One	7
2.2.1	Deriving Constants λ_1 , λ_2 , and Compressibility Factor Z_C	7
2.2.2	Deriving $\left(\frac{\partial P}{\partial V}\right)_T = 0$	7
2.2.3	Deriving $\left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0$	10
2.2.4	Deriving Constants	12
2.3	Problem A Part Two	13
2.4	Problem A Part Three	16
2.5	Problem A Part Four and Five	16
2.6	Problem A Part 6	20
2.6.1	The molar enthalpy in $\frac{kJ}{mol}$	20
2.6.1.1	Solving for ρ , density	21
2.6.2	Return to Molar Enthalpy	22
2.6.3	the molar entropy in $\frac{kJ}{kmolK}$	23
2.6.3.1	the fugacity coefficient in atm	23
2.6.4	Return to molar entropy	23
2.6.5	the thermal expansion coefficient in $\frac{1}{K}$	24
2.7	the molar heat capacity under constant volume and pressure in $\frac{kJ}{kmolK}$	25
3	Conclusion	25
4	References	26

1 Abstract

This report analyzes nitrogen using an exponential equation of state as an improvement to the Van der Waals equation of state. The theoretical constants λ_1 and λ_2 were first derived from the thermodynamic critical-point conditions $\left(\frac{\partial P}{\partial V}\right)_T = 0$ and $\left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0$, which also allowed the critical compressibility factor to be estimated. A second set of constants was then obtained by nonlinear regression using experimental nitrogen data.

The exponential equation of state was then rewritten in terms of compressibility factor, density, and temperature so that expressions for the primary residual properties could be developed. From this, equations were derived for the fugacity coefficient, residual enthalpy, and residual entropy, followed by derivations for the secondary thermal properties including the thermal expansion coefficient, heat capacity at constant volume, heat capacity at constant pressure, Joule–Thomson coefficient, and speed of sound. The required derivatives of the equation of state with respect to temperature and density were also obtained to support these calculations. Finally, the derived equations were applied to nitrogen at 10°C and 830 bar.

2 Problem A

Problem A asks a variety of questions related to an exponential equation of state as an improvement to the Van der Waals equation of state.

1. Using the proper thermodynamic and mathematical arguments estimate the numerical values of the constants λ_1 and λ_2 and the critical compressibility factor.
2. Perform a nonlinear regression and calculate the constants λ_1 and λ_2 using the experimental data for Nitrogen given in the table above.
3. Comment with a brief paragraph on whether the values calculated from the nonlinear regression are thermodynamically consistent
4. Derive all the formulas for the calculation of the primary residual thermal properties, fugacity coefficient, enthalpy and entropy.
5. Derive all the formulas for the calculation of the secondary thermal properties in the following order: thermal expansion coefficient, heat capacity under constant volume, heat capacity under constant pressure, Joule-Thomson coefficient and speed of sound
6. Calculate at temperature 10°C and pressure 830 bar
 - (a) The molar enthalpy in $\frac{\text{kJ}}{\text{mol}}$
 - (b) The molar entropy in $\frac{\text{kJ}}{\text{kmolK}}$
 - (c) The thermal expansion coefficient in $\frac{1}{\text{K}}$
 - (d) The molar heat capacity C_V in $\frac{\text{kJ}}{\text{kmolK}}$
 - (e) The molar heat capacity C_P in $\frac{\text{kJ}}{\text{kmolK}}$
 - (f) The Joule-Thomson coefficient in $\frac{\text{K}}{\text{bar}}$
 - (g) The speed of sounds in $\frac{\text{m}}{\text{s}}$
7. Prepare a table with the following columns
 - (a) Column 1: Property description
 - (b) Column 3: Value of property from Assignment 03
 - (c) Column 4: Value of property using the Peng-Robinson Equation of State
 - (d) Column 5: Value of property using the exponential EOS with theoretical values of λ_1 and λ_2
 - (e) Column 6: Value of property using the exponential EOS with numerical values of λ_1 and λ_2
 - (f) Column 7: NIST value from the table below

This problem relies on data obtained from Assignment 03. Specifically, the experimental data of nitrogen and the interpolation for values of 10°C and pressure 830 bar. The work has been added below, but isn't necessary to be read for this assignment.

2.1 Derivation of Nitrogen Data and Tables

The initial step to solving for these values was to import the given data into a CHE Calculator excel file. The data has been listed below

In order to continue with the question, I interpolated the values to determine the $Z = \text{compressibility factor}$ for both the column for 10°C and the row for 830 bar. There is an initial step that I needed to do before the Interpolation, and that was to convert the units for pressure. The data given in the chart is in terms of atm and the given pressure is in bar.

$$P(\text{bar}) = 830 \quad \therefore \quad P(\text{atm}) = 819.15$$

I was able to convert this value directly in excel through the CHE calculator's command "=BAR-toATM(input)" where input is the cell containing the pressure in bar.

P (atm)	Temperature (deg C)				
	0	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	1.000	1.000	1.001
5	0.998	0.999	1.000	1.001	1.001
10	0.996	0.998	1.000	1.001	1.002
30	0.959	0.996	1.001	1.005	1.007
50	0.958	0.996	1.004	1.009	1.014
100	0.984	1.004	1.018	1.027	1.033
200	1.036	1.057	1.072	1.082	1.089
300	1.132	1.146	1.154	1.160	1.163
400	1.256	1.254	1.253	1.250	1.247
600	1.524	1.495	1.471	1.450	1.432
800	1.798	1.723	1.697	1.658	1.625
1000	2.067	1.988	1.923	1.867	1.819

Figure 1: Initial Table values

Using the equation below, I can interpolate the Z.

$$y = y_1 + \frac{x - x_1}{x_2 - x_1} * (y_2 - y_1)$$

Applying the equation in terms of this questions yields,

$$Z = Z_1 + \frac{T - T_1}{T_2 - T_1} * (Z_2 - Z_1) \quad (1)$$

	Compressibility Factor					
	Temperature					
	0	10	25	50	75	100
Pressure (atm)	273.15	283.15	298.15	323.15	348.15	373.15
0	1.00000		1.00000	1.00000	1.00000	1.00000
1	1.00000		1.00000	1.00000	1.00000	1.00000
3	0.99900		0.99900	1.00000	1.00000	1.00100
5	0.99800		0.99900	1.00000	1.00100	1.00100
10	0.99600		0.99800	1.00000	1.00100	1.00200
30	0.95900		0.99600	1.00100	1.00500	1.00700
50	0.95800		0.99600	1.00400	1.00900	1.01400
100	0.98400		1.00400	1.01800	1.02700	1.03300
200	1.03600		1.05700	1.07200	1.08200	1.08900
300	1.13200		1.14600	1.15400	1.16000	1.16300
400	1.25600		1.25400	1.25300	1.25000	1.24700
600	1.52400		1.49500	1.47100	1.45000	1.43200
800	1.79800		1.72300	1.69700	1.65800	1.62500
819.15						
1000	2.06700		1.98800	1.92300	1.86700	1.81900

Figure 2: Parted data table before interpolation

I applied Equation One in the above chart down the column for 10°C, which gives the Z values corresponding to this constant temperature and varying pressures. This defines the equation variables as

Z = compressibility factor for 10°C

Z_1 = compressibility factor for 0°C

Z_2 = compressibility factor for 25°C

T = compressibility factor for 10°C

T_1 = compressibility factor for 0°C

T_2 = compressibility factor for 25°C

Next, I needed to interpolate the Z values across varying temperatures with a constant pressure of 819.15 bar. In this case the equation will be rewritten below as

$$Z = Z_1 + \frac{P - P_1}{P_2 - P_1} * (Z_2 - Z_1) \quad (2)$$

Z = compressibility factor for 819.15 atm

Z_1 = compressibility factor for 800 atm

Z_2 = compressibility factor for 1000 atm

P = compressibility factor for 819.15 atm

P_1 = compressibility factor for 800 atm

P_2 = compressibility factor for 1000 atm

Plugging these formulas in the CHE Calculator excel table gives the following tabulated data for this question, where the highlighted cell represents the Z value associated with 10°C and 819.15 atm

	Compressability Factor					
	Temperature					
	0	10	25	50	75	100
Pressure (atm)	273.15	283.15	298.15	323.15	348.15	373.15
0	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
1	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
3	0.99900	0.99900	0.99900	1.00000	1.00000	1.00100
5	0.99800	0.99840	0.99900	1.00000	1.00100	1.00100
10	0.99600	0.99680	0.99800	1.00000	1.00100	1.00200
30	0.95900	0.97380	0.99600	1.00100	1.00500	1.00700
50	0.95800	0.97320	0.99600	1.00400	1.00900	1.01400
100	0.98400	0.99200	1.00400	1.01800	1.02700	1.03300
200	1.03600	1.04440	1.05700	1.07200	1.08200	1.08900
300	1.13200	1.13760	1.14600	1.15400	1.16000	1.16300
400	1.25600	1.25520	1.25400	1.25300	1.25000	1.24700
600	1.52400	1.51240	1.49500	1.47100	1.45000	1.43200
800	1.79800	1.76800	1.72300	1.69700	1.65800	1.62500
819.15	1.82375	1.79360	1.74837	1.71864	1.67801	1.64357
1000	2.06700	2.03540	1.98800	1.92300	1.86700	1.81900

Figure 3: Interpolated Compression Factor Table

The next step is to transform the data table above to be in terms of ρ = density instead of Z = compressibility factor. I started with the definition of Z , where

$$Z = \frac{P}{\rho RT}$$

P = Pressure in atm

ρ = density in $\frac{\text{kmol}}{\text{m}^3}$

T = Temperature in K

$$R = \text{Gas Constant} = 0.082057 \frac{\text{L} * \text{atm}}{\text{mol} * \text{K}}$$

Rearranging, we find the expression

$$\rho = \frac{P}{ZRT}$$

It is important to note that the given pressure of 830 bar was converted to atm, but the temperature was kept as celcius. In the following transform, the temperature will be converted to kelvin, this is done by the calculation below. This is in an effort to simplify further calculations and to not require future unit conversions.

$$K(\text{kelvin}) = T^\circ\text{C} + 273.15$$

We can apply this equation to a new table, with the same axis of pressure and temperature. For an example,

$$\rho_1 = \frac{P_1}{Z_1 RT_1}$$

where ρ_1 is corresponding to the same temperature and pressure as the Z value is associated with. This gives the following chart of tabulated values. The units of density in this chart is in terms of $\frac{\text{mol}}{\text{L}}$, this will be converted to $\frac{\text{kmol}}{\text{m}^3}$.

$$1 \frac{\text{mol}}{\text{L}} * \frac{1000\text{L}}{1\text{m}^3} * \frac{1\text{kmol}}{1000\text{mol}} = 1 \frac{\text{kmol}}{\text{m}^3}$$

Therefore, there is no need for a unit conversion in the calculations as the ratio is 1:1 conversion. So, we can treat the table output as density in terms of $\frac{kmol}{m^3}$

	Density					
	Temperature					
	0	10	25	50	75	100
Pressure (atm)	273.15	283.15	298.15	323.15	348.15	373.15
0	0.000	0.000	0.000	0.000	0.000	0.000
1	0.045	0.043	0.041	0.038	0.035	0.033
3	0.134	0.129	0.123	0.113	0.105	0.098
5	0.224	0.216	0.205	0.189	0.175	0.163
10	0.448	0.432	0.410	0.377	0.350	0.326
30	1.396	1.326	1.231	1.130	1.045	0.973
50	2.329	2.211	2.052	1.878	1.735	1.610
100	4.534	4.339	4.071	3.705	3.408	3.162
200	8.613	8.242	7.734	7.036	6.470	5.998
300	11.824	11.350	10.700	9.804	9.053	8.424
400	14.209	13.716	13.038	12.039	11.201	10.476
600	17.565	17.075	16.404	15.382	14.484	13.684
800	19.851	19.475	18.978	17.778	16.890	16.078
819.15	20.039	19.656	19.150	17.975	17.088	16.277
1000	21.585	21.146	20.560	19.611	18.749	17.954

Figure 4: Density Table

Now that the data tables have been derived, I can begin to solve Problem A

2.2 Problem A, Part One

2.2.1 Deriving Constants λ_1 , λ_2 , and Compressibility Factor Z_C

For the first part of Problem A, I need to solve for the constants using thermodynamic and mathematical relations. I can solve for these values by knowing the principles under critical pressure and temperature. Under these conditions, the following derivatives are equal to zero.

$$\left(\frac{\partial P}{\partial V}\right)_T = 0$$

$$\left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0$$

Starting with the initial exponential Equation of State

$$\frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-\frac{a}{VRT}} \quad (3)$$

where the following are defined as

$$y = \frac{b}{4V}$$

$$a = \lambda_1 \frac{(RT_C)^2}{P_C}$$

$$b = \lambda_2 \frac{RT_C}{P_C}$$

2.2.2 Deriving $\left(\frac{\partial P}{\partial V}\right)_T = 0$

From here, I had to rearrange Equation Three in terms of just pressure.

$$P = \frac{RT}{V} \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-\frac{a}{VRT}} \quad (4)$$

In order to simplify the derivation, I set the function of y to variable x

$$x = \frac{1 + y + y^2 - y^3}{(1 - y)^3}$$

This will now rearrange Equation Four into

$$P = \frac{RT}{V} x e^{-\frac{a}{VRT}} \quad (5)$$

In order to further simplify, I took the natural log of both sides.

$$\ln P = \ln(RT) + \ln(x) - \ln(V) - \frac{a}{VRT}$$

Now, taking the derivative with respect to Volume over a constant temperature,

$$\begin{aligned} \left(\frac{\partial \ln(P)}{\partial V} \right)_T &= \left(\frac{\partial}{\partial V} \right) \left[\ln(RT) + \ln(x) - \ln(V) - \frac{a}{VRT} \right] \\ \left(\frac{\partial \ln(P)}{\partial V} \right)_T &= 0 + \frac{\partial \ln(x)}{\partial V} - \frac{\partial \ln V}{\partial V} - \frac{\partial}{\partial V} \left(\frac{a}{VRT} \right) \end{aligned}$$

This can be simplified by applying the chain rule (example shown below),

$$\left(\frac{\partial \ln(P)}{\partial V} \right)_T = \frac{\partial}{\partial V} (\ln(P)) = \frac{1}{P} \frac{\partial P}{\partial V}$$

And now applying the above principle to the x and V

$$\frac{1}{P} \left(\frac{\partial P}{\partial V} \right)_T = \frac{1}{x} \frac{\partial x}{\partial V} - \frac{1}{V} + \frac{a}{V^2 RT}$$

Starting with the definition of x as a function of y and y as a function of V,

$$\left(\frac{\partial P}{\partial V} \right)_T = P \left[\frac{1}{x} * \frac{\partial x}{\partial y} * \frac{\partial y}{\partial V} - \frac{1}{V} + \frac{a}{V^2 RT} \right] \quad (6)$$

I then found the derivative of x with respect to y using the quotient rule

$$\begin{aligned} x &= \frac{1 + y + y^2 - y^3}{(1 - y)^3} \\ x' = \frac{\partial x}{\partial y} &= \frac{(1 - y)^3 [1 + 2y - 3y^2] - (1 + y + y^2 - y^3) [-3(1 - y)^2]}{((1 - y)^3)^2} \\ &= \frac{(1 - y)^3 (1 + 2y - 3y^2) + 3(1 - y)^2 (1 + y + y^2 - y^3)}{(1 - y)^6} \\ &= \frac{(1 - y)(1 + 2y - 3y^2) + 3(1 + y + y^2 - y^3)}{(1 - y)^4} \\ &= \frac{1 - y + 2y - 2y^2 - 3y^2 + 3y^3 + 3 + 3y + 3y^2 - 3y^3}{(1 - y)^4} \\ x' = \frac{\partial x}{\partial y} &= \frac{-2y^2 + 4y + 4}{(1 - y)^4} \quad (7) \end{aligned}$$

From here, I now move to the second derivative of x,

$$\begin{aligned} x'' = \frac{\partial^2 x}{\partial y^2} &= \frac{(1 - y)^4 [-4y + 4] - (-2y^2 + 4y + 4) [4(1 - y)^3 * -1]}{((1 - y)^4)^2} \\ &= \frac{(1 - y)^4 (-4y + 4) + 4(-2y^2 + 4y + 4)(1 - y)^3}{(1 - y)^8} \\ &= \frac{(1 - y)(-4y + 4) - 8y^2 + 16y + 16}{(1 - y)^5} \\ &= \frac{-4y + 4 + 4y^2 - 4y - 8y^2 + 16y + 16}{(1 - y)^5} \end{aligned}$$

$$= \frac{-4y^2 + 8y + 20}{(1-y)^5}$$

Then, I solved for the derivative of y with respect to V , and also solved for its second derivative

$$y = \frac{b}{4V}$$

$$y' = \left(\frac{\partial y}{\partial V} \right)_T = -\frac{b}{4V^2}$$

which can be simplified from the definition of y to be

$$y' = \left(\frac{\partial y}{\partial V} \right)_T = -\frac{y}{V} \quad (8)$$

$$y'' = \left(\frac{\partial^2 y}{\partial V^2} \right)_T = \frac{b}{2V^3} \quad (9)$$

Returning to equation 6, and plugging in equations 7 and 8

$$\left(\frac{\partial P}{\partial V} \right)_T = P \left[\frac{x'y'}{x} - \frac{1}{V} + \frac{a}{V^2 RT} \right] \quad (10)$$

Simplifying here by plugging the definition of y'

$$\left(\frac{\partial P}{\partial V} \right)_T = P \left[-\frac{x'y}{xV} - \frac{1}{V} + \frac{a}{V^2 RT} \right]$$

Then pulling out $\frac{1}{V}$

$$\left(\frac{\partial P}{\partial V} \right)_T = \frac{P}{V} \left[-\frac{x'y}{x} - 1 + \frac{a}{VRT} \right]$$

Under critical conditions this equation is equal to zero so

$$\left(\frac{\partial P}{\partial V} \right)_T = \frac{P_C}{V_C} \left[-\frac{x'y_C}{x} - 1 + \frac{a}{V_C RT_C} \right] = 0$$

$$-\frac{x'y_C}{x} - 1 + \frac{a}{V_C RT_C} = 0$$

The next step was to simplify the variable of a . First, I plugged in the definition of " a "

$$\frac{a}{V_C RT_C} = \frac{\lambda_1 \frac{(RT_C)^2}{P_C}}{V_C RT_C} = \lambda_1 \frac{RT_C}{V_C P_C}$$

Recalling the definition of " b " and plugging in

$$b = \lambda_2 \frac{RT_C}{P_C}$$

$$\therefore \frac{b}{\lambda_2} = \frac{RT_C}{P_C}$$

$$\frac{a}{VRT_C} = \lambda_1 \frac{1}{V_C} \frac{b}{\lambda_2}$$

And now knowing that $y = \frac{b}{4V}$ and setting $k = 4\frac{\lambda_1}{\lambda_2}$

$$\frac{a}{V_C RT_C} = \frac{\lambda_1}{\lambda_2} \frac{b}{V} = k y_C$$

$$\therefore -\frac{x'y_C}{x} - 1 + k y_C = 0$$

$$-x'y_C - x + kxy_C = 0$$

$$kxy_C = x + x'y_C$$

$$k = \frac{1}{y_C} + \frac{x'}{x} \quad (11)$$

2.2.3 Deriving $\left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0$

Now, onto solving for the second derivative, starting with equation 10

$$\begin{aligned} \left(\frac{\partial P}{\partial V}\right)_T &= P \left[\frac{x'y'}{x} - \frac{1}{V} + \frac{a}{V^2 RT} \right] \\ \left(\frac{\partial^2 P}{\partial V^2}\right)_T &= P \left[\frac{x'y'}{x} - \frac{1}{V} + \frac{a}{V^2 RT} \right]' + \left(\frac{\partial P}{\partial V}\right)_T \left[\frac{x'y'}{x} - \frac{1}{V} + \frac{a}{V^2 RT} \right] \end{aligned}$$

Here I know every part of the equation except $\left[\frac{x'y'}{x} - \frac{1}{V} + \frac{a}{V^2 RT} \right]'$. However, as mentioned in the previous section, under critical conditions the $\left(\frac{\partial P}{\partial V}\right)_T = 0$. So that part of the equation cancels.

$$\therefore \left(\frac{\partial^2 P}{\partial V^2}\right)_T = P \left[\frac{x'y'}{x} - \frac{1}{V} + \frac{a}{V^2 RT} \right]' \quad (12)$$

Moving on to the derivative,

$$\begin{aligned} \left[\frac{x'y'}{x} - \frac{1}{V} + \frac{a}{V^2 RT} \right]' &= \left[\frac{x'y'}{x} \right]' - \left[\frac{1}{V} \right]' + \left[\frac{a}{V^2 RT} \right]' \\ \left[\frac{x'y'}{x} - \frac{1}{V} + \frac{a}{V^2 RT} \right]' &= \frac{1}{V^2} - \frac{2a}{RTV^3} + \left[\frac{x'y'}{x} \right]' \end{aligned}$$

I now need to solve for the final derivative $\left[\frac{x'y'}{x} \right]'$. By plugging in the alternate definition of y'

$$\frac{x'y'}{x} = -\frac{y}{V} \frac{x'}{x}$$

Now this gives the following equation,

$$\left[\frac{x'y'}{x} - \frac{1}{V} + \frac{a}{V^2 RT} \right]' = \frac{1}{V^2} - \frac{2a}{RTV^3} - \left[\frac{y}{V} \frac{x'}{x} \right]' \quad (13)$$

This would then make the derivative now equal to

$$\left[\frac{y}{V} \frac{x'}{x} \right]' = \left[\left[\frac{y}{V} \right] * \left[\frac{x'}{x} \right]' + \left[\frac{y}{V} \right]' * \left[\frac{x'}{x} \right] \right] \quad (14)$$

Now, $\frac{y}{V}$ is usable as is so I then moved on to solve for $\left[\frac{x'}{x} \right]'$

$$\left[\frac{x'}{x} \right]' = \frac{\partial}{\partial y} \left[\frac{x'}{x} \right] * \frac{\partial y}{\partial V}$$

$$\frac{\partial}{\partial y} \left[\frac{x'}{x} \right] = \frac{x * x'' - x' * x'}{x^2} = \frac{x * x'' - (x')^2}{x^2} = \frac{x''}{x} - \left(\frac{x'}{x} \right)^2$$

$$\frac{\partial y}{\partial V} = -\frac{y}{V}$$

$$\therefore \left[\frac{x'}{x} \right]' = \left[\frac{x''}{x} - \left(\frac{x'}{x} \right)^2 \right] * \left(-\frac{y}{V} \right)$$

Moving onto the next derivative $\left[\frac{y}{V}\right]'$

$$\left[\frac{y}{V}\right]' = \frac{\partial}{\partial V} \left[\frac{y}{V}\right] = \frac{\partial}{\partial V} (yV^{-1}) = y * [-V^{-2}] + V^{-1} * y' = \frac{y}{V^2} - \frac{y}{V^2} = -\frac{2y}{V^2}$$

Plugging all the above equations into Equation 14 yields

$$\left[\frac{y}{V} \frac{x'}{x}\right]' = \left[-\left(\frac{y}{V}\right)^2 \left[\frac{x''}{x} - \left(\frac{x'}{x}\right)^2\right] - \frac{2y}{V^2} \frac{x'}{x}\right]$$

I can then take this expression and plug it into Equation 13,

$$\left[\frac{x'y'}{x} - \frac{1}{V} - \frac{a}{V^2 RT}\right]' = \frac{1}{V^2} - \frac{2a}{RTV^3} + \left[-\left(\frac{y}{V}\right)^2 \left[\frac{x''}{x} - \left(\frac{x'}{x}\right)^2\right] - \frac{2y}{V^2} \frac{x'}{x}\right]$$

Now taking this equation and plugging into Equation 12 gives,

$$\left(\frac{\partial^2 P}{\partial V^2}\right)_T = P \left[\frac{1}{V^2} - \frac{2a}{RTV^3} - \left(\frac{y}{V}\right)^2 \left[\frac{x''}{x} - \left(\frac{x'}{x}\right)^2\right] - \frac{2y}{V^2} \frac{x'}{x}\right]$$

Now taking this equation under critical conditions gives

$$\left(\frac{\partial^2 P}{\partial V^2}\right)_T = P_C \left[\frac{1}{V_C^2} - \frac{2a}{RT_C V_C^3} - \left(\frac{y}{V_C}\right)^2 \left[\frac{x''}{x} - \left(\frac{x'}{x}\right)^2\right] - \frac{2y}{V_C^2} \frac{x'}{x}\right] = 0$$

$$\frac{1}{V_C^2} - \frac{2a}{RT_C V_C^3} - \frac{y_C^2}{V_C^2} \left[\frac{x''}{x} - \left(\frac{x'}{x}\right)^2\right] - \frac{2y_C}{V_C^2} \frac{x'}{x} = 0$$

And now multiplying everything by V_C^2

$$1 - \frac{2a}{RT_C V_C} - y_C^2 \left[\frac{x''}{x} - \left(\frac{x'}{x}\right)^2\right] - 2y_C \frac{x'}{x} = 0$$

Once again, simplifying "a" where $a = \lambda_1 \frac{(RT_C)^2}{P_C}$

$$\frac{2a}{RT_C V_C} = \frac{2 * \lambda_1 \frac{(RT_C)^2}{P_C}}{RT_C V_C} = 2\lambda_1 \frac{RT_C}{P_C V_C}$$

recalling that $\frac{b}{\lambda_2} = \frac{RT_C}{P_C}$ and $y = \frac{b}{4V}$ which then means $4y = \frac{b}{V}$

$$\frac{2a}{RT_C V_C} = 2\lambda_1 \frac{1}{V_C} \frac{b}{\lambda_2} = 2 \frac{\lambda_1}{\lambda_2} \frac{b}{V_C} = 2 \frac{\lambda_1}{\lambda_2} * 4y_C$$

Here I recall that I set $k = 4 \frac{\lambda_1}{\lambda_2}$

$$\frac{2a}{RT_C V_C} = 2ky_C$$

Plugging this back in to the original equation gives,

$$1 - 2ky_C + y_C^2 \left[\frac{x''}{x} - \left(\frac{x'}{x}\right)^2\right] + 2y_C \frac{x'}{x} = 0 \quad (15)$$

2.2.4 Deriving Constants

Here I recall both critical equations solved for, Equations 11 and 15

$$k = \frac{1}{y_C} + \frac{x'}{x}$$

$$1 - 2ky_C + y_C^2 \left[\frac{x''}{x} - \left(\frac{x'}{x} \right)^2 \right] + 2y_C \frac{x'}{x} = 0$$

Rearranging Equation 11 gives,

$$k = \frac{1}{y_C} + \frac{x'}{x}$$

$$ky_C = 1 + \frac{x'}{x} y_C$$

And now plugging this into Equation 15

$$1 - 2\left(1 + \frac{x'}{x} y_C\right) + y_C^2 \left[\frac{x''}{x} - \left(\frac{x'}{x} \right)^2 \right] + 2y_C \frac{x'}{x} = 0$$

$$1 - 2 - 2\frac{x'}{x} y_C + y_C^2 \left[\frac{x''}{x} - \left(\frac{x'}{x} \right)^2 \right] + 2y_C \frac{x'}{x} = 0$$

$$-1 + y_C^2 \left[\frac{x''}{x} - \left(\frac{x'}{x} \right)^2 \right] = 0$$

$$y_C^2 \left[\frac{x''}{x} - \left(\frac{x'}{x} \right)^2 \right] = 1$$

From here I can solve for y_C by entering this equation into excel and using goal seek. Both x and x' have been derived earlier in terms of y , here I used those equations by entering y_C .

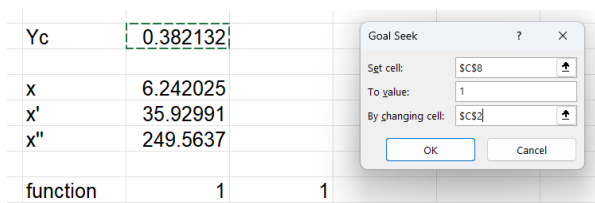


Figure 5: y_C Goal Seek

This gives a value of

$$y_C = 0.382132$$

From here, I can recall Equation 11 to solve for the k value,

$$k = \frac{1}{y_C} + \frac{x'}{x}$$

I can use the earlier values of x' and x along with the solved for value of y_C

This returns a value of $k = 8.373028782$. From here I returned to the definition of k I established.

$$k = 4 \frac{\lambda_1}{\lambda_2}$$

$$\lambda_1 = \frac{k\lambda_2}{4}$$

In order to continue solving, I returned to equation 3.

$$\frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-\frac{a}{VKT}}$$

From here we can simplify by applying

$$x = \frac{1 + y + y^2 - y^3}{(1 - y)^3}$$

$$\frac{a}{V_C RT_C} = \frac{\lambda_1}{\lambda_2} \frac{b}{V} = k y_C$$

This then gives the following equation, where x , k , and y_C are known

$$Z_C = x e^{-k y_C}$$

$$Z_C = 6.242024885 * e^{-(8.373028782)(0.382131899)} = 0.254540134$$

Now, I have values for x, x', x'', Z_C, y_C , I can solve for both λ_1 and λ_2 . Starting with the definitions of y and b .

$$y = \frac{b}{4V} \quad \therefore \quad y_C = \frac{b}{4V_C} \quad b = \lambda_2 \frac{RT_C}{P_C}$$

$$4y_C V_C = b = \lambda_2 \frac{RT_C}{P_C}$$

$$4y_C V_C = \lambda_2 \frac{RT_C}{P_C}$$

$$4y_C = \frac{\lambda_2}{Z_C}$$

$$\lambda_2 = 4y_C Z_C$$

$$\lambda_2 = 4 * (0.382131899) * (0.254540134) = 0.389071619$$

Now returning to the λ_1 definition,

$$\lambda_1 = \frac{k \lambda_2}{4} = \frac{(8.373028782) * (0.389071619)}{4} = 0.814426965$$

Below I have summarized all the variables solved for

Problem A Part One Table Summary

Variable	Equation	Value
x	$x = \frac{1+y+y^2-y^3}{(1-y)^3}$	6.242024885
x'	$x' = \frac{-2y^2+4y+4}{(1-y)^4}$	35.92991489
x''	$x'' = \frac{-4y^2+8y+20}{(1-y)^5}$	249.5636517
y_C	$y_C^2 \left[\frac{x''}{x} - \left(\frac{x'}{x} \right)^2 \right] = 1$	0.382131899
Z_C	$Z_C = x e^{-k y_C}$	0.254540134
λ_1	$\lambda_1 = \frac{k \lambda_2}{4}$	0.814426965
λ_2	$\lambda_2 = 4 y_C Z_C$	0.389071619

2.3 Problem A Part Two

Part Two asks to solve for λ_1 and λ_2 using nonlinear regression and the experimental data. In order to begin this regression, I started with rearranging the EOS.

$$\frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-\frac{a}{V RT}}$$

$$y = \frac{b}{4V}$$

$$b = \lambda_2 \frac{RT_C}{P_C}$$

Starting with the equation in terms of Z,

$$Z = \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-\frac{a}{VRT}}$$

the next step is to rearrange the exponential factor,

$$\begin{aligned} -\frac{a}{VRT} \quad a &= \lambda_1 \frac{(RT_C)^2}{P_C} \\ \therefore -\frac{a}{VRT} &= -\frac{\lambda_1 \frac{(RT_C)^2}{P_C}}{VRT} = -\frac{\lambda_1 (RT_C)^2}{VRT P_C} \end{aligned}$$

Next I defined the reduced pressure and temperature, and z

$$\begin{aligned} P_r &= \frac{P}{P_C} \quad T_r = \frac{T}{T_C} \quad Z = \frac{PV}{RT} \\ -\frac{a}{VRT} &= -\frac{\lambda_1 P_r}{T_r^2 Z} \end{aligned}$$

The next step is to redefine y, where

$$\begin{aligned} y &= \frac{b}{4V} \quad b = \lambda_2 \frac{RT_C}{P_C} \\ y &= \frac{b}{4V} = \frac{\lambda_2 \frac{RT_C}{P_C}}{4V} \end{aligned}$$

Starting from the Z definition,

$$Z = \frac{PV}{RT} \quad \therefore \quad V = \frac{ZRT}{P}$$

So y now becomes,

$$y = \frac{\lambda_2 \frac{RT_C}{P_C}}{4 \frac{ZRT}{P}} = \frac{\lambda_2 P_r}{4 T_r Z}$$

This now brings the EOS to

$$Z = \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-\frac{\lambda_1 P_r}{Z T_r^2}} \quad \text{where} \quad y = \frac{\lambda_2 P_r}{4 T_r Z}$$

Now, I took the equation for Z and defined some function of Z

$$\begin{aligned} f(Z) &= Z - RHS \\ \text{where} \quad RHS &= \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-\frac{\lambda_1 P_r}{Z T_r^2}} \end{aligned}$$

From here I will then use solver to minimize the value of f(Z) to as close to zero as possible by altering the values of λ_1 and λ_2 .

In order to reduce complications and errors when typing, I broke the EOS into 5 tables

1. Table One is the interpolated nitrogen data
2. Table Two is the corresponding y values for each temperature and pressure
3. Table 3 represents the value of the exponent, $-\frac{\lambda_1 P_r}{Z T_r^2}$
4. Table 4 is the calculated RHS
5. Table 5 finally represents the Z from table one minus the values of RHS from table 4

The next step was to sum all of the values of table 5. I used values $\lambda_1 = 0.8$ and $\lambda_2 = 0.2$ as initial guesses to guide the solver. Solver was then used to reduce this summed value to zero, or as close as it could find. Below is the five tables and summation provided.

Pressure (atm)	Compressability Factor						
	Temperature						
	0	10	25	50	75	100	
	273.15	283.15	298.15	323.15	348.15	373.15	
0	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	
1	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	
3	0.99900	0.99900	0.99900	1.00000	1.00000	1.00100	
5	0.99800	0.99840	0.99900	1.00000	1.00100	1.00100	
10	0.99600	0.99680	0.99800	1.00000	1.00100	1.00200	
30	0.95900	0.97380	0.99600	1.00100	1.00500	1.00700	
50	0.95800	0.97320	0.99600	1.00400	1.00900	1.01400	
100	0.98400	0.99200	1.00400	1.01800	1.02700	1.03300	
200	1.03600	1.04440	1.05700	1.07200	1.08200	1.08900	
300	1.13200	1.13760	1.14600	1.15400	1.16000	1.16300	
400	1.25600	1.25520	1.25400	1.25300	1.25000	1.24700	
600	1.52400	1.51240	1.49500	1.47100	1.45000	1.43200	
800	1.79800	1.76800	1.72300	1.69700	1.65800	1.62500	
819.15	1.82375	1.79360	1.74837	1.71864	1.67801	1.64357	
1000	2.06700	2.03540	1.98800	1.92300	1.86700	1.81900	

Figure 6: Table of Compressability Factor Values

Pressure (atm)	y						
	Temperature						
	0	10	25	50	75	100	
	273.15	283.15	298.15	323.15	348.15	373.15	
0	0	0	0	0	0	0	
1	0.001151962	0.001111278	0.001055369	0.000973722	0.000903801	0.000843249	
3	0.003459345	0.003337171	0.003169278	0.002921167	0.002711404	0.002527222	
5	0.005771352	0.005565295	0.005282129	0.004868612	0.004514492	0.004212033	
10	0.011565882	0.011148456	0.010574844	0.009737224	0.009028983	0.008415959	
30	0.036036347	0.034235308	0.031788235	0.029182489	0.02697914	0.025121621	
50	0.060123272	0.057094024	0.052980392	0.048492151	0.044786978	0.041580329	
100	0.117069298	0.112024001	0.105116475	0.095650529	0.088004012	0.081631081	
200	0.222386465	0.212806987	0.199691469	0.181664625	0.167061221	0.154866679	
300	0.305290253	0.293058567	0.276274715	0.25313407	0.233741691	0.217519106	
400	0.366866844	0.354135783	0.336640961	0.310845136	0.289216386	0.270488875	
600	0.453528303	0.440866736	0.42355963	0.397167528	0.373986706	0.353316649	
800	0.512552565	0.502840763	0.490014817	0.459032358	0.436092259	0.415138003	
819.15	0.517408859	0.507526787	0.494461998	0.464101423	0.441204971	0.420270239	
1000	0.557311025	0.54597528	0.530869925	0.506355896	0.484092772	0.463578377	

Figure 7: Table of y Values

Pressure (atm)	exponent						
	Temperature						
	0	10	25	50	75	100	
	273.15	283.15	298.15	323.15	348.15	373.15	
0	0	0	0	0	0	0	
1	-0.00515	-0.00479	-0.004323818	-0.00368	-0.00317	-0.00276	
3	-0.01547	-0.0144	-0.012984438	-0.01104	-0.00951	-0.00827	
5	-0.02581	-0.02401	-0.021640729	-0.0184	-0.01584	-0.01379	
10	-0.05172	-0.04809	-0.043324827	-0.03681	-0.03168	-0.02755	
30	-0.16115	-0.14769	-0.130235474	-0.11031	-0.09466	-0.08224	
50	-0.26887	-0.2463	-0.217059123	-0.1833	-0.15714	-0.13611	
100	-0.52353	-0.48327	-0.430659137	-0.36156	-0.30877	-0.26722	
200	-0.9945	-0.91805	-0.81813013	-0.6867	-0.58615	-0.50696	
300	-1.36524	-1.26426	-1.13188946	-0.95685	-0.8201	-0.71205	
400	-1.64061	-1.52775	-1.37920821	-1.175	-1.01474	-0.88545	
600	-2.02815	-1.9019	-1.735311467	-1.5013	-1.31216	-1.15659	
800	-2.29211	-2.16926	-2.007576431	-1.73515	-1.53006	-1.35896	
819.15	-2.31382	-2.18948	-2.025796402	-1.75431	-1.548	-1.37576	
1000	-2.49226	-2.35534	-2.17495862	-1.91403	-1.69848	-1.51753	

Figure 8: Table of Exponent Values

Pressure (atm)	RHS						
	Temperature						
	0	10	25	50	75	100	
	273.15	283.15	298.15	323.15	348.15	373.15	
0	1	1	1	1	1	1	
1	0.999459	0.999654	0.9999	1.000216	1.000446	1.000614	
3	0.998393	0.998975	0.999713	1.00066	1.001348	1.00185	
5	0.997346	0.998316	0.999544	1.001119	1.002262	1.0031	
10	0.994822	0.996752	0.999198	1.002334	1.00461	1.006275	
30	0.985666	0.991605	0.99892	1.008141	1.01481	1.019694	
50	0.978961	0.988555	1.000393	1.015423	1.026309	1.034195	
100	0.971884	0.989369	1.011449	1.039697	1.060145	1.075042	
200	0.991891	1.021939	1.060205	1.109907	1.146198	1.172582	
300	1.043292	1.08227	1.132535	1.199475	1.248607	1.284765	
400	1.109171	1.156053	1.217727	1.299645	1.361439	1.4067	
600	1.262565	1.323339	1.405829	1.517919	1.603972	1.667969	
800	1.431438	1.514315	1.632825	1.752486	1.863534	1.946843	
819.15	1.448672	1.532244	1.65174	1.775793	1.889208	1.974434	
1000	1.616024	1.704113	1.830043	2.002029	2.138547	2.242689	

Figure 9: Table of RHS Values

		f(Z)								
		Temperature								
		0	10	25	50	75	100			
Pressure (atm)		273.15	283.15	296.15	323.15	348.15	373.15		Sum	
0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000		0.00000	
1	0.00054	0.00035	0.00010	-0.00022	-0.00045	-0.00061	-0.00061		-0.00029	
3	0.00061	0.00003	-0.00071	-0.00066	-0.00135	-0.00085	-0.00085		-0.00294	
5	0.00065	0.00008	-0.00054	-0.00112	-0.00126	-0.00210	-0.00210		-0.00429	
10	0.00118	0.00005	-0.00120	-0.00233	-0.00361	-0.00427	-0.00427		-0.01019	
30	-0.02667	-0.01781	-0.00292	-0.00714	-0.00981	-0.01269	-0.01269		-0.07704	
50	-0.02096	-0.01535	-0.00439	-0.01142	-0.01731	-0.02020	-0.02020		-0.08964	
100	0.01212	0.00283	-0.00745	-0.02170	-0.03315	-0.04204	-0.04204		-0.08959	
200	0.04411	0.02246	-0.00320	-0.03791	-0.06420	-0.08358	-0.08358		-0.12232	
300	0.08871	0.05533	0.01347	-0.04548	-0.08861	-0.12177	-0.08834		-0.09834	
400	0.14683	0.09915	0.03627	-0.04665	-0.11144	-0.15970	-0.03554		-0.03554	
600	0.26144	0.18906	0.08917	-0.04692	-0.15397	-0.23597	0.10281		0.10281	
800	0.36656	0.25369	0.09018	-0.05549	-0.20553	-0.32184	0.12756		0.12756	
819.15	0.37508	0.26135	0.09663	-0.05716	-0.21120	-0.33086	0.13384		0.13384	
1000	0.45098	0.33129	0.15796	-0.07903	-0.27155	-0.42369	0.16596		0.16596	
									0.0000005196	Total

Figure 10: Table of $f(Z) = Z - \text{RHS Values}$

2.4 Problem A Part Three

Part three asks me to comment on the thermodynamic consistency of the regression values. While the values for λ_1 and λ_2 are similar, they are made under different assumptions.

When solving for the values using theory, I assumed a specific state at the critical point. This assumption led to the derivatives always equaling zero. The Regressed values are based on empirical data across various pressures and temperatures. Since these were solved for by data, there should be consistent correlation. But there is nothing keeping the values strictly thermodynamically consistent, so no. They are not thermodynamically consistent, even if they are similar values.

2.5 Problem A Part Four and Five

In Part four of this assignment, I am asked to derive the formulas for the calculation of the primary residual thermal properties, fugacity coefficient, enthalpy and entropy. Then, for part five I am asked to derive the formulas for the calculation of the secondary thermal properties in the following order: thermal expansion coefficient, heat capacity under constant volume, heat capacity under constant pressure, Joule-Thomson coefficient and speed of sound.

In order to start solving for these values, I need to rewrite the EOS in terms of $Z(\rho, T)$. Starting from the beginning with Equation 3,

$$\frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-\frac{a}{VRT}}$$

This can be rewritten where $Z = \frac{PV}{RT}$ and $\rho = \frac{1}{V}$. This can then make $y = \frac{b\rho}{4V} = \frac{b\rho}{4}$.

$$Z = \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-\frac{a\rho}{RT}}$$

From here, I set $n = \frac{b}{4}$, so now $y = n\rho$. Now plugging this in I get the equation

$$Z(\rho, T) = \frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} e^{-\frac{a\rho}{RT}}$$

Here I have a function of Z in terms of ρ and T. Now I need Z specifically in terms of A and B variables. I define A and B as

$$A = \frac{aP}{(RT)^2} \quad B = \frac{bP}{RT}$$

Turning to the exponential now,

$$-\frac{a}{VRT} \quad \text{where} \quad V = \frac{ZRT}{P}$$

$$-\frac{a}{VRT} = -\frac{aP}{Z(RT)^2} = -\frac{A}{Z}$$

Then I looked to B,

$$y = \frac{b}{4V} \quad \therefore \quad b = 4yV$$

$$B = \frac{bP}{RT} = \frac{4VyP}{RT} = 4Zy$$

$$\therefore \quad Z = \frac{B}{4y}$$

Combining this with the exponential, I get

$$-\frac{a}{VRT} = -4\frac{A}{B}y$$

And now returning to the original Z function gives

$$Z = \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-4\frac{A}{B}y}$$

Now, I have equations for Z as both a function of ρ, T and A, B

$$Z(A, B) = \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-4\frac{A}{B}y}$$

$$Z(\rho, T) = \frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} e^{-\frac{a\rho}{RT}}$$

From chapter four of the book, I have the following equations for the thermal properties. If I define the following C_1 and C_2 , then

$$C_1 = Z + T \left(\frac{\partial Z}{\partial T} \right)_\rho$$

$$C_2 = Z + \rho \left(\frac{\partial Z}{\partial \rho} \right)_T$$

$$\ln \phi = (Z - 1) - \ln Z + \int_0^\rho (Z - 1) \frac{d\rho}{\rho}$$

$$\frac{H^R}{RT} = (Z - 1) - \int_0^\rho T \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho}$$

$$\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi$$

$$\beta = \frac{1}{T} \frac{C_1}{C_2}$$

$$C_v^R = \int_\infty^V T \left(\frac{\partial^2 P}{\partial T^2} \right)_V dV$$

$$C_p - C_v = R \frac{C_1^2}{C_2}$$

$$C_p = (C_p - C_v) + C_v$$

$$\mu_{JT} = \frac{\beta T - 1}{\rho C_p}$$

$$\nu^2 = \frac{RT}{M_W} \frac{C_p}{C_v} C_2$$

With variables C_1 and C_2 , all the functions above are functions of 3 derivatives.

$$\left(\frac{\partial Z}{\partial T}\right)_\rho \quad \left(\frac{\partial Z}{\partial \rho}\right)_T \quad \left(\frac{\partial^2 P}{\partial T^2}\right)_V$$

First, I need to solve for $\left(\frac{\partial Z}{\partial T}\right)_\rho$

$$Z(\rho, T) = \frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} e^{-\frac{a\rho}{RT}}$$

$$\left(\frac{\partial Z}{\partial T}\right)_\rho = \left[\frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} \right] * e^{-\frac{a\rho}{RT}} * \left[\frac{a\rho}{RT^2} \right]$$

$$\left(\frac{\partial Z}{\partial T}\right)_\rho = \frac{a\rho Z}{RT^2}$$

Now moving onto the next derivative, $\left(\frac{\partial Z}{\partial \rho}\right)_T$

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = \left[\frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} \right] \left[e^{-\frac{a\rho}{RT}} \right]' + \left[\frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} \right]' \left[e^{-\frac{a\rho}{RT}} \right]$$

$$\left[e^{-\frac{a\rho}{RT}} \right]' = e^{-\frac{a\rho}{RT}} * \left[-\frac{a}{RT} \right]$$

$$x = \frac{1 + y + y^2 - y^3}{(1 - y)^3} \approx \left[\frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} \right]$$

$$\therefore x' = \left[\frac{1 + y + y^2 - y^3}{(1 - y)^3} \right]' = \frac{-2y^2 + 4y + 4}{(1 - y)^4} \approx \frac{-2n^2\rho^2 + 4n\rho + 4}{(1 - n\rho)^4} * n = \frac{\partial x}{\partial y} \frac{\partial y}{\partial \rho}$$

Now taking all these values and plugging in yields,

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = \left[\frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} \right] \left[e^{-\frac{a\rho}{RT}} * \left[-\frac{a}{RT} \right] \right] + \left[\frac{-2n^2\rho^2 + 4n\rho + 4}{(1 - n\rho)^4} \right] \left[e^{-\frac{a\rho}{RT}} \right]$$

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = \left[-\frac{a}{RT} \left[\frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} \right] + \frac{-2n^2\rho^2 + 4n\rho + 4}{(1 - n\rho)^4} \right] e^{-\frac{a\rho}{RT}}$$

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = \left[-\frac{a}{RT} x(\rho) + x'(\rho) \right] e^{-\frac{a\rho}{RT}}$$

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = \left[-\frac{a}{RT} + \frac{x'(\rho)}{x(\rho)} \right] x(\rho) e^{-\frac{a\rho}{RT}}$$

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = Z \left[\frac{x'(\rho)}{x(\rho)} - \frac{a}{RT} \right]$$

The final derivative I need to solve for is $\left(\frac{\partial^2 P}{\partial T^2}\right)_V$. This requires me to return to Equation 4,

$$P = \frac{RT}{V} \frac{1 + y + y^2 - y^3}{(1 - y)^3} e^{-\frac{a}{VKT}}$$

$$x = \frac{1 + y + y^2 - y^3}{(1 - y)^3}$$

This will now rearrange Equation Four into

$$P = \frac{RT}{V} x e^{-\frac{a}{VKT}} \quad (16)$$

In order to further simplify, I took the natural log of both sides.

$$\ln P = \ln(RT) + \ln(x) - \ln(V) - \frac{a}{VRT}$$

This follows a similar process as part one of this assignment,

$$\left(\frac{d \ln P}{dT}\right)_V = \frac{d}{dT} \left(\ln(RT) + \ln(x) - \ln(V) - \frac{a}{VRT} \right)_V$$

The left hand side can be simplified as

$$\left(\frac{\partial \ln(P)}{\partial T}\right)_V = \frac{\partial}{\partial T} (\ln(P)) = \frac{1}{P} \frac{\partial P}{\partial T}$$

So the initial equation can be written as

$$\frac{1}{P} \left(\frac{\partial P}{\partial T}\right)_V = \frac{d}{dT} \left(\ln(RT) + \ln(x) - \ln(V) - \frac{a}{VRT} \right)_V$$

Since V is being held constant, the $\ln(V) = 0$. Also, $\ln(x)$ isn't a function of temperature so that becomes zero.

$$\frac{1}{P} \left(\frac{\partial P}{\partial T}\right)_V = \frac{1}{T} + \frac{a}{VRT^2}$$

$$\left(\frac{\partial P}{\partial T}\right)_V = P \left[\frac{1}{T} + \frac{a}{VRT^2} \right]$$

From here the first derivative is solved for, now I need to take the second derivative with respect to T over constant volume.

$$\left(\frac{\partial^2 P}{\partial T^2}\right)_V = P \left[\frac{1}{T} + \frac{a}{VRT^2} \right]' + P' \left[\frac{1}{T} + \frac{a}{VRT^2} \right]$$

$$\left[\frac{1}{T} + \frac{a}{VRT^2} \right]' = -\frac{1}{T^2} - \frac{2a}{VRT^3}$$

$$P' = \left(\frac{\partial P}{\partial T}\right)_V = P \left[\frac{1}{T} + \frac{a}{VRT^2} \right]$$

$$\left(\frac{\partial^2 P}{\partial T^2}\right)_V = P \left[-\frac{1}{T^2} - \frac{2a}{VRT^3} \right] + P \left[\frac{1}{T} + \frac{a}{VRT^2} \right]^2$$

$$\left(\frac{\partial^2 P}{\partial T^2}\right)_V = P \left[-\frac{1}{T^2} - \frac{2a}{VRT^3} + \left[\frac{1}{T} + \frac{a}{VRT^2} \right]^2 \right]$$

This can be further simplified by expanding the square,

$$\left(\frac{1}{T} + \frac{a}{VRT^2} \right)^2 = \frac{1}{T^2} + \frac{2a}{VRT^3} + \frac{a^2}{V^2 R^2 T^4}$$

Now substituting this back in,

$$\left(\frac{\partial^2 P}{\partial T^2}\right)_V = P \left[-\frac{1}{T^2} - \frac{2a}{VRT^3} + \frac{1}{T^2} + \frac{2a}{VRT^3} + \frac{a^2}{V^2 R^2 T^4} \right]$$

The first four terms cancel, leaving

$$\left(\frac{\partial^2 P}{\partial T^2}\right)_V = P \frac{a^2}{V^2 R^2 T^4}$$

From here I can now simplify the expressions for C_1 and C_2 ,

$$C_1 = Z + T \left(\frac{\partial Z}{\partial T} \right)_\rho$$

$$C_1 = Z + T \left[\frac{a \rho Z}{RT^2} \right]$$

$$C_1 = Z \left(1 + \frac{a\rho}{RT} \right)$$

And now for C_2

$$C_2 = Z + \rho \left(\frac{\partial Z}{\partial \rho} \right)_T$$

$$C_2 = Z + \rho Z \left[\frac{x'(\rho)}{x(\rho)} - \frac{a}{RT} \right]$$

With all these derivatives solved and equations written, the last step is to solve for these values at a specific temperature and pressure.

2.6 Problem A Part 6

For part 6, I need to solve for the above properties at a given temperature and pressure.

- Calculate at temperature 10°C and pressure 830 bar

1. The molar enthalpy in $\frac{\text{kJ}}{\text{mol}}$
2. The molar entropy in $\frac{\text{kJ}}{\text{kmolK}}$
3. The thermal expansion coefficient in $\frac{1}{\text{K}}$
4. The molar heat capacity C_V in $\frac{\text{kJ}}{\text{kmolK}}$
5. The molar heat capacity C_P in $\frac{\text{kJ}}{\text{kmolK}}$
6. The Joule-Thomson coefficient in $\frac{\text{K}}{\text{bar}}$
7. The speed of sounds in $\frac{\text{m}}{\text{s}}$

2.6.1 The molar enthalpy in $\frac{\text{kJ}}{\text{mol}}$

In order to solve for the enthalpy, I can start with the residual definition,

$$H = H^R + H^{IG}$$

$$H^{IG} = \text{"HIG(comp1,T=10}^\circ\text{C)"}} = -437.7625975 \frac{\text{kJ}}{\text{kmol}}$$

$$\frac{H^R}{RT} = (Z - 1) - \int_0^\rho T \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho}$$

And now plugging in $\left(\frac{\partial Z}{\partial T} \right)_\rho$,

$$\left(\frac{\partial Z}{\partial T} \right)_\rho = \frac{a\rho Z}{RT^2}$$

$$\frac{H^R}{RT} = (Z - 1) - \int_0^\rho T \frac{a\rho Z}{RT^2} \frac{d\rho}{\rho}$$

$$\frac{H^R}{RT} = (Z - 1) - \int_0^\rho \frac{aZ}{RT} d\rho$$

2.6.1.1 Solving for ρ , density

In order to solve for the density, I need to return to Z as a function of ρ and T . At $T = 10^\circ C = 283.15\text{ K}$ and $P = 830\text{ bar}$, the density is found by solving

$$Z(\rho, T) = \frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} e^{-\frac{a\rho}{RT}}$$

$$\frac{P}{\rho RT} = \frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} \exp\left(-\frac{a\rho}{RT}\right)$$

where

$$a = \lambda_1 \frac{(RT_C)^2}{P_C} \quad n = \frac{b}{4} \quad b = \lambda_2 \frac{RT_C}{P_C}$$

In order to solve further I know that

$$T_C = -146.95^\circ C = 126.2\text{ K} \quad P_C = 33.5\text{ atm}$$

And then recalling the theoretical values for λ_1 and λ_2

$$a = 2.67637, \quad n = \frac{b}{4} = 0.0304645, \quad R = 0.082057 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}$$

I then set the following equations and variables into excel,

$$LHS = \frac{P}{\rho RT}$$

$$RHS = \frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3} \exp\left(-\frac{a\rho}{RT}\right)$$

$$Balance = LHS - RHS = 0$$

By inputting the above equations, I was able to goal seek the value of density, ρ .

$$\rho = 16.97297738 \text{ kmol/m}^3$$

Then, using

$$Z = \frac{P}{\rho RT}$$

gives

$$Z = \frac{819.1463114\text{ atm}}{(16.97297738 \frac{\text{kmol}}{\text{m}^3})(0.082057 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}})(283.15\text{ K})}$$

$$Z = \frac{819.1463114\text{ atm}}{(16.97297738 \frac{\text{kmol}}{\text{m}^3})(0.082057 \frac{\text{m}^3 \cdot \text{atm}}{\text{kmol} \cdot \text{K}})(283.15\text{ K})} = 2.077166197$$

The excel solution has been added below,

	Density	
balance	2.29602E-08	
LHS	2.077166197	
RHS	2.077166174	
RHS top	1.637867767	
RHS bottom	0.117405818	
RHS exp	-1.904509524	
a	2.607097766	
b	0.120270671	
n	0.030067668	
lambda1	0.814426965	
lambda2	0.389071619	
R	0.082057	L*atm/mol K
Tc	-146.95	C
	126.2	K
Pc	33.5	atm
P	830	bar
	819.1463114	atm
T	10	C
	283.15	k
Density rho	16.97297738	mol/L
	16.97297738	kmol/m^3
Z	2.077166197	
RT	23.23443955	

Figure 11: Caption

2.6.2 Return to Molar Enthalpy

Now that the density has been solved for, I returned to the residual enthalpy

$$\frac{H^R}{RT} = (Z - 1) - \int_0^P \frac{aZ}{RT} d\rho$$

$$\frac{H^R}{RT} = (Z - 1) - \frac{a}{RT} \int_0^P Z d\rho$$

$$H^R = RT(Z - 1) - a \int_0^P Z d\rho$$

This was solved for through numerical estimation. I created a column of density values from 0 to 75 in 0.1 increments. I then solved for the corresponding Z value from these ρ values. Next was to use the fukima function.

$$\int_0^P Z d\rho = " = fAkima(\Delta\rho, \Delta Z, 5 = integral, 16.97297738, 0)" = 1.004644487$$

From here the equation becomes

$$H^R = RT(Z - 1) - a(1.004644487)$$

where

$$a = 2.67637, \quad T = 283.15K \quad R = 0.082057 \frac{L * atm}{mol * K} \quad Z = 2.077166197$$

$$\therefore H^R = (0.082057 \frac{L * atm}{mol * K})(283.15K)(2.077166197 - 1) - (2.67637)(1.004644487) = 22.40814649 kJ/kmol$$

$$H^{IG} = -437.7625975 \frac{kJ}{kmol} = -0.437762598 \frac{kJ}{mol}$$

$$H^R = 22.40814649 \frac{kJ}{kmol} = 0.02240814649 \frac{kJ}{mol}$$

$$H = H^{IG} + H^R = -415.354451 \frac{kJ}{kmol} = -0.415354451 \frac{kJ}{mol}$$

2.6.3 the molar entropy in $\frac{kJ}{kmolK}$

Earlier, I derived the formula for S^R as

$$\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi$$

2.6.3.1 the fugacity coefficient in atm

Now I need to solve for $\ln \phi$

$$\ln \phi = (Z - 1) - \ln Z + \int_0^P (Z - 1) \frac{d\rho}{\rho}$$

$$\ln \phi = (Z - 1) - \ln Z + \int_0^P \frac{(Z - 1)}{\rho} d\rho$$

This integral was solved similarly to the enthalpy integral. I created one column for Z, one for (Z-1), and then one for $\frac{(Z-1)}{\rho}$. I once again used the fukima function.

$$\int_0^P (Z - 1) \frac{d\rho}{\rho} = "fAkima(\Delta\rho, \Delta \frac{(Z - 1)}{\rho}, 5, 16.97297738, 0)" = 0.00867996$$

$$\ln \phi = (Z - 1) - \ln Z + 0.00867996$$

where $Z = 2.077166197$

$$\ln \phi = (2.077166197 - 1) - \ln(2.077166197) + 0.00867996$$

$$\therefore \ln \phi = 0.354841598 \quad \therefore \phi = e^{0.354841598} = 1.425954762 atm$$

2.6.4 Return to molar entropy

$$\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi$$

Now i recall the values of R, H^R , T, and $\ln \phi$

$$R = 0.082057 \frac{m^3 * atm}{kmol * K} \quad H^R = 22.40814649 \frac{kJ}{kmol} \quad T = 283.15K \quad \ln \phi = 0.354841598$$

$$S^R = \frac{H^R}{T} - R \ln \phi \quad \therefore S^R = 0.050021546 \frac{kJ}{kmolK}$$

$$S^{IG} = " = SIG(comp1, T = 10^\circ C, P = 830 bar) = -57.28208751 \frac{kJ}{kmolK}$$

$$S = S^R + S^{IG} = 0.050021546 - 57.28208751 = -57.23206596 \frac{kJ}{kmolK}$$

2.6.5 the thermal expansion coefficient in $\frac{1}{K}$

Earlier, I derived the formal for β

$$\beta = \frac{1}{T} \frac{C_1}{C_2}$$

where

$$C_1 = Z + T \left(\frac{\partial Z}{\partial T} \right)_\rho$$

$$C_1 = 2.077166197 + T \left(\frac{\partial Z}{\partial T} \right)_\rho$$

$$\left(\frac{\partial Z}{\partial T} \right)_\rho = \frac{a\rho Z}{RT^2}$$

$$C_1 = Z + T \left(\frac{a\rho Z}{RT^2} \right)$$

$$C_1 = Z + \frac{a\rho Z}{RT}$$

$$a = 2.67637, \quad T = 283.15K \quad R = 0.082057 \frac{L * atm}{mol * K} \quad Z = 2.077166197 \quad \rho = 16.97297738 \text{ kmol/m}^3$$

$$\therefore C_1 = 6.033149002$$

Now moving onto C_2

$$C_2 = Z + \rho \left(\frac{\partial Z}{\partial \rho} \right)_T$$

$$\left(\frac{\partial Z}{\partial \rho} \right)_T = Z \left[\frac{x'(\rho)}{x(\rho)} - \frac{a}{RT} \right]$$

$$C_2 = Z + \rho Z \left[\frac{x'(\rho)}{x(\rho)} - \frac{a}{RT} \right]$$

$$x = \frac{1 + n\rho + n^2\rho^2 - n^3\rho^3}{(1 - n\rho)^3}$$

$$x'(\rho) = \frac{-2n^2\rho^2 + 4n\rho + 4}{(1 - n\rho)^4} * n$$

Evaluating these two function at $n = 0.030067668$ and $\rho = 16.97297738 \frac{\text{kmol}}{\text{m}^3}$

$$x = 13.95048218$$

$$x' = 2.887280699$$

$$\therefore C_2 = 5.417926688$$

And now returning to β

$$\beta = \frac{1}{T} \frac{C_1}{C_2}$$

$$\beta = \frac{1}{283.15} * \frac{6.033149002}{5.417926688} = 0.003932732 \frac{1}{K}$$

2.7 the molar heat capacity under constant volume and pressure in $\frac{kJ}{kmolK}$

In order to solve for the heat capacities, C_v and C_P , I started by solving for the residual heat capacity under constant volume. Then using the CVIG function I can calculate the $C_v = C_V^{IG} + C_v^R$

$$C_v^R = \int_{\infty}^V T \left(\frac{\partial^2 P}{\partial T^2} \right)_V dV$$

Earlier, I already solved for the derivative,

$$\left(\frac{\partial^2 P}{\partial T^2} \right)_V = P \frac{a^2}{V^2 R^2 T^4}$$

$$C_v^R = \int_{\infty}^V T P \frac{a^2}{V^2 R^2 T^4} dV$$

$$C_v^R = \int_{\infty}^V T P \frac{a^2}{V^2 R^2 T^4} dV$$

And knowing that $P = \frac{ZRT}{V}$

$$C_v^R = \int_{\infty}^V T \frac{ZRT}{V} \frac{a^2}{V^2 R^2 T^4} dV$$

$$C_v^R = \frac{a^2}{RT^2} \int_{\infty}^V \frac{Z}{V^3} dV$$

We can derive this relation starting with

$$\rho V = 1$$

$$\rho dV + V d\rho = 0$$

$$\rho dV = -V d\rho$$

$$\frac{dV}{V} = -\frac{d\rho}{\rho}$$

$$dV = -\frac{d\rho}{\rho^2}$$

And now plugging in yields

$$C_v^R = -\frac{a^2}{RT^2} \int_0^{\rho} \rho Z d\rho = -\frac{a^2}{RT^2} * 0.503147445 = -0.001033156$$

$$CV^{IG} = CV^{IG}(comp1, 10^\circ C) = 20.87745872$$

$$C_v = C_v^R + C_v^{IG} = -0.001033156 + 20.87745872 = 20.87642557 \frac{kJ}{kmolK}$$

3 Conclusion

In this assignment, nitrogen was analyzed using an exponential equation of state and compared with both the Van der Waals and Peng–Robinson equations of state. The theoretical constants λ_1 and λ_2 were first estimated from the critical-point conditions by applying the requirements $\left(\frac{\partial P}{\partial V} \right)_T = 0$ and $\left(\frac{\partial^2 P}{\partial V^2} \right)_T = 0$. This provided theoretical values for the equation of state parameters as well as an estimate for the critical compressibility factor. A second set of parameters was then obtained through nonlinear regression using experimental nitrogen data, allowing a comparison between thermodynamically derived constants and empirically fitted constants.

The exponential equation of state was then rewritten in terms of compressibility factor, density, and temperature so that the required thermodynamic derivatives could be derived. From these results, expressions were developed for the fugacity coefficient, residual enthalpy, residual entropy, thermal expansion coefficient, heat capacities, Joule–Thomson coefficient, and speed of sound. This showed that once the equation of state is placed in a usable $Z(\rho, T)$ form, both the primary and secondary thermal properties can be obtained systematically from a small set of derivatives and integral relations.

4 References

- ChatGPT was used on the development of structure and readability of this report. It was also used to help guide the thermodynamic derivations and to correct the work in this report.
- Chapters One through Four of the Class Textbook were used to help solve this assignment numerically